

MoACB-WSF: Multi-Objective Automated CNN-BiLSTM for Wind Speed Forecasting

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Abstract—Accurate wind speed forecasting (WSF) is essential for integrating renewable wind energy into power grids and optimizing generation scheduling. However, developing high-performance deep learning models currently relies on manual, expert-driven configuration, making it difficult to balance prediction accuracy with the model complexity required for edge deployment. To address this challenge, this paper proposes a novel framework, namely Multi-Objective Automated CNN-BiLSTM for Wind Speed Forecasting (MoACB-WSF). This approach formulates the automated generation of hybrid neural networks as a bi-objective minimization problem, concurrently optimizing forecasting error and parameter efficiency. Utilizing a unified fixed-length hybrid encoding scheme, alongside customized crossover and mutation operators, MoACB-WSF enables the end-to-end joint optimization of the entire model structure and training settings via NSGA-II. Extensive evaluations across three real-world wind farm datasets demonstrate that the framework consistently outperforms state-of-the-art manually designed models and a leading neuroevolution baseline. Validated through multiple independent runs, MoACB-WSF exhibits strong robustness, autonomously discovering stable, lightweight forecasting models that achieve an optimal accuracy-complexity trade-off for practical wind farm systems.

Index Terms—Wind speed forecasting, Automated deep learning, Multi-objective evolutionary algorithm, Hybrid neural networks.

I. INTRODUCTION

AS a core pillar of the global renewable energy mix and a cornerstone of the worldwide low-carbon energy transition, wind power has solidified its position as a dominant mainstream source in the global clean energy landscape. As documented in the Global Wind Energy Council (GWEC) Global Wind Report 2025, the global wind energy sector added 117 GW of new installed capacity in 2024, marking its continued expansion into emerging geographic markets while further reinforcing its central role in underpinning global energy security and sustainable development [1]. However, the inherent intermittency, randomness, and non-stationarity of wind speed introduce profound uncertainty into wind power output, which has emerged as a key bottleneck limiting the efficient, high-value utilization of wind energy resources. Large deviations in wind speed forecasting directly result in suboptimal generation scheduling, elevated wind curtailment, diminished wind energy utilization efficiency, and weakened

economic viability of wind farm operations [2]. Therefore, accurate and reliable wind speed forecasting is an indispensable prerequisite for addressing these challenges, and is critically important for unlocking the full potential of wind energy resources, enhancing the economic viability of wind farms, and supporting the continued large-scale deployment of wind power worldwide.

For high-precision wind speed forecasting (WSF), deep learning (DL) has firmly established itself as both the mainstream technical approach and the state-of-the-art methodological paradigm in recent years [3]–[6]. With superior nonlinear temporal modeling and hierarchical feature extraction capabilities, DL methods effectively capture the inherent intermittency, randomness, and non-stationarity of wind speed series, and have been extensively validated to deliver superior forecasting accuracy and robustness across diverse onshore and offshore wind farm scenarios. In particular, hybrid DL architectures that combine convolutional neural networks (CNNs) for local feature extraction and bidirectional long short-term memory (BiLSTM) networks for long-range temporal dependency modeling consistently outperform single-structure DL counterparts in WSF tasks. Nevertheless, the development of high-performance DL forecasting models remains predominantly reliant on manual neural architecture design and hyperparameter tuning. This manual paradigm not only requires extensive domain expertise and time-consuming trial-and-error experimentation, but also fails to effectively resolve the inherent trade-off between prediction accuracy and model complexity. Manually designed models typically suffer from either excessive parameterization that impedes real-time inference on resource-constrained edge devices in wind farms, or suboptimal forecasting accuracy due to oversimplified structural configurations.

To address these fundamental limitations, automated deep learning (ADL) has emerged as a transformative technical paradigm that automates the end-to-end design of neural network architectures and their associated training configurations with minimal human intervention [7]. By leveraging evolutionary algorithms, which excel at global optimization in large and complex hybrid search spaces, ADL enables the simultaneous automatic discovery of optimized neural architectures and training hyperparameters for deep learning models. This capability has achieved significant success across diverse research domains: in computer vision, evolutionary ADL frameworks have evolved architectures for image classification, object detection, and semantic segmentation that consistently outperform manually designed state-of-the-art models on standard benchmarks including ImageNet and

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COCO [7]–[10]; and in industrial Internet of Things (IIoT) applications, ADL has facilitated the rapid development of high-performance task-specific models for real-time equipment monitoring, anomaly detection, and predictive maintenance, which have been successfully deployed in practical industrial scenarios [11]–[13]. Despite these widespread successes across multiple domains, the application of ADL in WSF remains largely underexplored. Most existing ADL studies in WSF suffer from three critical limitations: first, they address neural architecture optimization and hyperparameter tuning separately, failing to exploit the inherent synergies between network structure and training configurations; second, they predominantly adopt a single-objective optimization perspective that solely focuses on prediction accuracy, neglecting model complexity which is essential for edge deployment; and third, they are primarily designed for single-structure DL models, lacking specialized support for hybrid CNN-BiLSTM architectures that have demonstrated superior performance in WSF tasks. This significant research gap motivates the development of a specialized multi-objective evolutionary ADL framework tailored for WSF, which can automatically discover high-performance, lightweight hybrid deep learning models with minimal manual intervention.

To fill this gap, this paper proposes a novel *Multi-Objective Automated CNN-BiLSTM for Wind Speed Forecasting (MoACB-WSF)* framework tailored specifically for high-precision and lightweight wind speed prediction, which is indispensable for wind farm real-time operation scheduling and power system frequency regulation. The proposed framework adopts an end-to-end multi-objective evolutionary pipeline based on the Non-dominated Sorting Genetic Algorithm II (NSGA-II), which automatically designs both the neural architectures and training parameters of hybrid CNN-BiLSTM models, enabling the generation of high-performance forecasting models without extensive manual intervention. The core contributions of this work are summarized as follows:

- (i) To address the limitations of existing WSF studies that rely on manual design, adopt single-structure architectures, and optimize solely prediction accuracy, this paper presents the first multi-objective automated deep learning framework for hybrid CNN-BiLSTM models in the WSF domain. It achieves the simultaneous joint optimization of neural architectures and training hyperparameters, effectively mitigating the inherent trade-off between prediction accuracy and model complexity.
- (ii) An efficient fixed-length hybrid encoding scheme is developed under the NSGA-II optimization framework, which can simultaneously encode the hybrid network topology, CNN/BiLSTM layer-wise structural parameters, and key training hyperparameters including learning rate, batch size, optimizer type, and dropout rate. This unified encoding scheme enables end-to-end joint optimization of the entire model configuration within a single search space, eliminating the need for laborious manual trial-and-error experimentation.
- (iii) Extensive experiments are conducted on three real-world wind farm datasets collected from diverse lo-

cations. Comparative evaluations with five state-of-the-art manually designed hybrid models and one leading neuroevolution-based forecasting approach demonstrate that the proposed MoACB-WSF consistently outperforms all baseline models across all standard evaluation metrics while maintaining significantly lower model complexity, achieving a superior accuracy-complexity trade-off suitable for edge deployment in practical wind farm systems.

II. RELATED WORKS

A. DL-Based WSF

Due to the powerful nonlinear temporal modeling capabilities of deep learning, a wide range of deep learning (DL) architectures have been developed for wind speed forecasting (WSF). Zhang et al. [14] proposed a spatiotemporal dual-UNet (STD-UNet) fully convolutional network for large-scale refined grid vector wind speed prediction, which achieves state-of-the-art accuracy while maintaining high inference efficiency. Liu et al. [6] further advanced temporal-spatial modeling by developing a Transformer-based WindTrans framework with a dynamic graph encoder, enabling the simultaneous capture of time-varying spatial correlations and long-range temporal dependencies for multi-site forecasting. To enhance physical consistency and environmental adaptability, Mi et al. [15] constructed a physics-informed TCN-TFT hybrid model integrated with quantile regression, where Bernoulli's principle and the ideal gas law are embedded into the loss function to generate physically consistent probabilistic predictions. Malakar et al. [16] designed an adaptive DL model for variable terrain conditions; it adopts PACF to reduce IMF dimensions and selects LSTM or bidirectional feature LSTM adaptively via sample entropy, thus achieving stable accuracy across plain and complex terrains. For probabilistic forecasting and limited-data scenarios, Saeed et al. [17] proposed an enhanced multi-quantile loss combined with multi-scale dilated causal convolutions, which simultaneously produces deterministic forecasts and high-quality prediction intervals with improved computational efficiency. Sun and Tian [18] developed a decomposition-ensemble transfer learning framework integrating VMD, sample entropy, and customized DL modules, effectively addressing the few-shot challenge in WSF. Additionally, Hu et al. [19] presented an OWT-STGradRAM spatiotemporal DL model, which encodes wind farm features as images and integrates orthogonal wind transformation and gradient-based activation mapping for ultra-short-term wind speed prediction.

Although deep learning (DL) models have achieved remarkable success in wind speed forecasting, their development still heavily relies on manual trial-and-error to tune hyperparameters and design neural architectures, requiring substantial expertise across both deep learning and wind energy engineering. This limitation becomes more severe when complex model ensembles are adopted, as the expanded parameter space and architectural choices can significantly degrade forecasting stability and performance. It remains challenging to manually identify the optimal combination of DL modules and their corresponding configurations, often leading to suboptimal

solutions. Furthermore, existing studies predominantly prioritize prediction accuracy while overlooking model complexity, which is a critical constraint for practical deployment on resource-limited wind farm edge devices.

B. ADL-Based Time Series Forecasting

Recent advances in automated deep learning (ADL) have significantly improved the design efficiency of time series forecasting models across diverse application domains, reducing reliance on manual architecture engineering. A representative line of work leverages neuroevolution for adaptive model construction: Lyu et al. [20] proposed ONE-NAS, an online neuroevolution-based NAS method that continuously evolves recurrent neural networks to handle non-stationary dynamics, demonstrating superior performance on wind turbine datasets. Building on hybrid architectural designs, Xie et al. [21] developed an evolutionary NAS approach that automatically synthesizes CNN-GRU-Attention models for time series prediction in turntable servo systems, while Cai et al. [22] introduced EST-NAS, which couples evolutionary strategies with gradient-based refinement to discover high-performance deep network topologies without manual intervention. Beyond single-objective accuracy maximization, Li et al. [23] formulated a multimodal multiobjective NAS framework that simultaneously optimizes prediction accuracy and model resilience, yielding lightweight architectures suitable for fault-tolerant deployment. Resource-constrained scenarios have also received attention: Chen et al. [24] integrated zero-shot NAS with quantization-aware training to automatically produce compact deep networks for lithium-ion battery fault diagnosis, Theodorakos et al. [25] proposed an island transpeciation co-evolutionary framework that coordinates multiple optimizers to design MIMO-NARX deep neural networks for country-scale air quality forecasting. Targeting multivariate temporal dependencies, Chen et al. [26] proposed a scale-aware NAS framework incorporating multi-scale decomposition and adaptive graph learning, enabling automated architecture discovery without domain-specific prior knowledge. In the context of renewable energy, Pujari et al. [27] applied evolutionary multi-objective NAS to wind forecasting, optimizing across LSTM, nonlinear autoregressive, and wavelet neural network families on real nonlinear wind time series data.

These studies demonstrate that ADL techniques can effectively discover high-performance time series forecasting models with minimal manual intervention, outperforming many manually designed state-of-the-art approaches. However, most existing ADL-based time series forecasting methods suffer from two critical limitations. First, they predominantly adopt a single-objective optimization perspective that solely focuses on maximizing prediction accuracy, neglecting model complexity which is essential for practical edge deployment. Though a few studies have attempted multi-objective optimization, they either only optimize hyper-parameters without neural architecture search, or restrict their scope to single-structure, omitting the joint optimization of advanced hybrid DL architectures. Second, almost all existing ADL frameworks are designed for general time series tasks, and there are very few

specialized ADL frameworks tailored specifically for wind speed forecasting. None of them have fully considered the unique characteristics of wind speed series, such as strong intermittency, randomness, and non-stationarity, which significantly limits their performance in wind farm applications.

III. THE PROPOSED METHOD

In this section, the technical design and implementation details of the MoACB-WSF framework are systematically elaborated.

A. Overview

Fig. 1 gives the overall framework of the proposed MoACB-WSF method. First, real supervisory control and data acquisition (SCADA)-based wind farm systems are used to gather the wind speed time series dataset. Then, the raw data is transformed into the standard time series dataset by data preprocessing and divided into the training set, validation set and test set. The proposed MoACB-WSF is performed on the training and validation datasets to find the optimized hybrid CNN-BiLSTM architectures and hyper-parameters by a customized NSGA-II. After the evolutionary process, the models in the final external archive are fully trained with related hyper-parameters for evaluation. The pseudo-code of MoACB-WSF is given in Algorithm 1. Firstly, generate an initialized parent population P_0 randomly by the proposed fixed-length hybrid encoding strategy, which means the neural architectures and hyperparameters for a fixed-length blocks-based hybrid CNN-BiLSTM model, and initialize the external archive A_E by an empty set. Secondly, generate an offspring population Q_t by the proposed customized crossover and mutation methods based on P_t , and evaluate the bi-objective fitness of the combined population $R_t = P_t \cup Q_t$ by Pareto-based strategy with non-dominated sorting. Thirdly, update the external archive A_E with the non-dominated solutions from R_t by the proposed archive updating method, and select the next-generation population P_{t+1} via elitist environmental selection. Repeat the second and third phase until the maximal generation time T . Finally, assess the model performance of wind speed forecasting based on the optimized hybrid CNN-BiLSTM with the non-dominated architectures and hyperparameters in A_E .

B. The Proposed Hybrid Encoding Strategy

The design of an effective encoding strategy is a fundamental challenge in neuroevolution-based neural architecture search (NAS), as it fundamentally governs the search space expressiveness, evolutionary operator validity, solution feasibility, and final model performance [28]. Existing encoding paradigms are mainly divided into variable-length and fixed-length types: the former allows dynamic adjustment of network blocks but suffers from inherent length drift during evolution, high probability of generating invalid solutions, and requires complex length-adaptive evolutionary operators [29], [30]; while the latter maintains constant individual dimensionality throughout the evolutionary process, simplifies crossover and

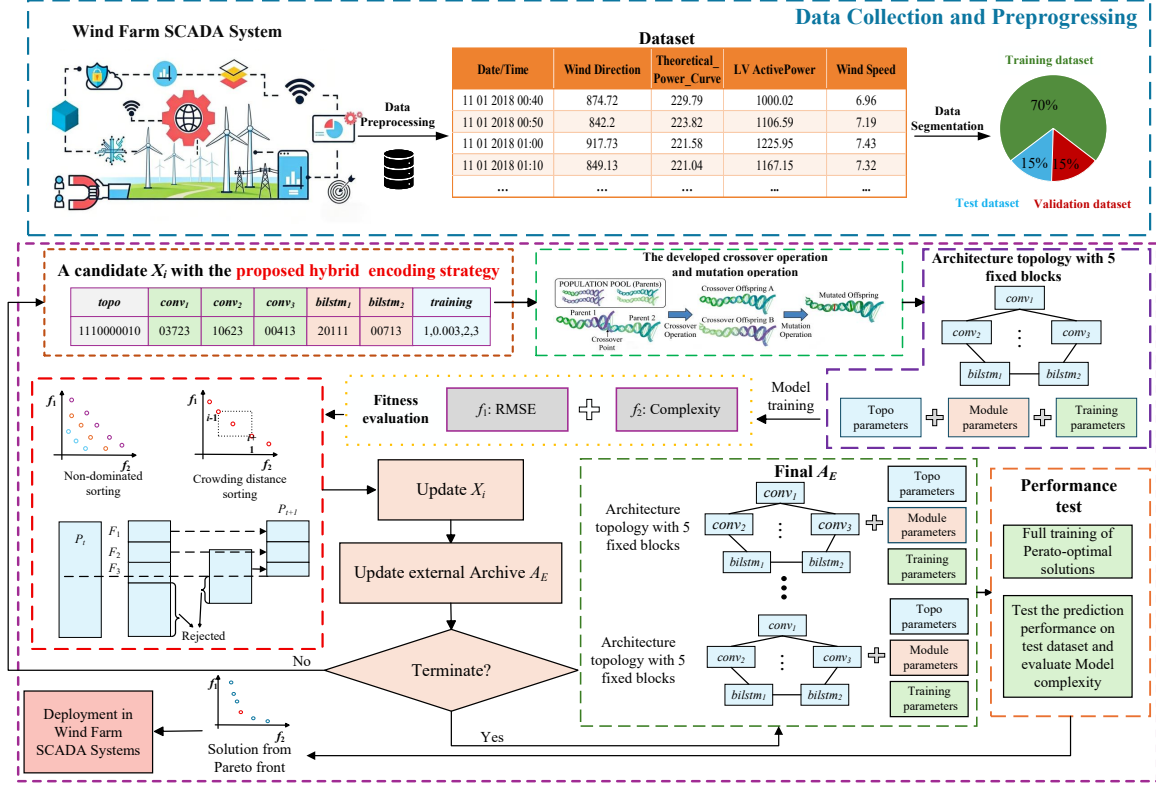


Fig. 1. The overall framework of the proposed MoACB-WSF.

mutation designs, eliminates invalid solutions at the encoding level, and exhibits superior search efficiency and stability, making it particularly suitable for time-sensitive and resource-constrained applications such as real-time wind speed forecasting. Furthermore, most existing NAS approaches only encode network topologies and module parameters while treating training hyperparameters as manually tuned constants, which inevitably leads to suboptimal solutions due to the synergistic interaction between architecture and training configuration [31]. Even the few evolutionary-based joint optimization studies adopt separate encoding schemes for discrete architectural parameters and continuous training parameters, compromising individual structural integrity and failing to fully exploit cross-parameter coupling relationships [32].

To address the above limitations, we propose a unified fixed-length 39-dimensional hybrid encoding scheme that seamlessly integrates discrete network topology, integer module parameters, and continuous training hyperparameters into a single mixed-variable vector, enabling end-to-end joint optimization of all critical components of the hybrid CNN-BiLSTM model. Specifically, we construct this encoding scheme for the hybrid CNN-BiLSTM backbone with 5 fixed independent blocks (3 CNN modules and 2 BiLSTM modules), which has been widely validated as an effective architecture for wind speed time series forecasting tasks. Each individual in the NSGA-II population is represented as a mixed-variable vector:

$$\mathbf{x} = [\mathbf{x}_{\text{topo}}, \mathbf{x}_{\text{CNN}}, \mathbf{x}_{\text{BiLSTM}}, \mathbf{x}_{\text{train}}] \quad (1)$$

where $\mathbf{x}_{\text{topo}}$ denotes the binary topology encoding segment, $\mathbf{x}_{\text{CNN}}$ represents the integer encoding for CNN module parameters, $\mathbf{x}_{\text{BiLSTM}}$ is the integer encoding for BiLSTM module parameters, and $\mathbf{x}_{\text{train}}$ contains the mixed-type encoding for training hyperparameters.

Fig. 2 presents the hierarchical structure of the proposed encoding strategy, which partitions the 39-dimensional vector into four functionally independent segments corresponding to different components of the forecasting model. Fig. 3 further illustrates the one-to-one mapping process, where a single mixed-variable vector is decoded into a complete hybrid CNN-BiLSTM model with its corresponding training configuration. The following subsections detail the encoding rules for each segment.

1) *Topology Encoding*: The first 10 dimensions are reserved for binary topology encoding, which defines the directed feedforward connections among the 5 core modules labeled sequentially as Module 1 to Module 5. The topology encoding is formulated as:

$$\begin{aligned} \mathbf{x}_{\text{topo}} &= [\text{bin}_1, \text{bin}_2, \dots, \text{bin}_{10}], \\ \text{bin}_i &= \text{Randint}(0, 1) \end{aligned} \quad (2)$$

The 10-bit binary codes are sequentially mapped to a 4×4 upper triangular adjacency matrix A to represent the network topology. For any element a_{ij} in matrix A , $a_{ij} = 1$ indicates a direct connection from Module i to Module $j + 1$, while $a_{ij} = 0$ means no such connection exists. A strict structural

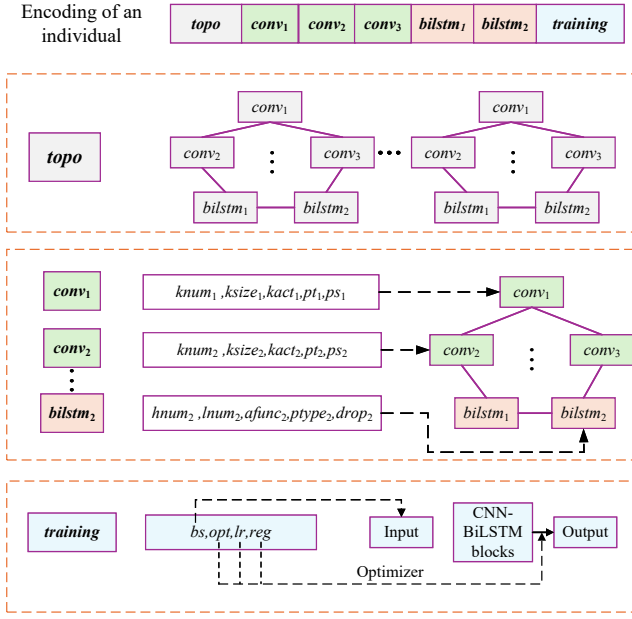


Fig. 2. Overall framework of the proposed hybrid fixed-length encoding strategy.

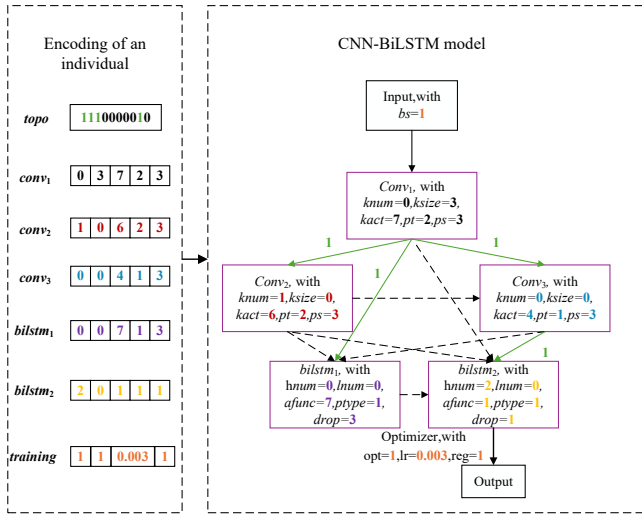


Fig. 3. Concrete encoding example of an initialized individual.

validity constraint is enforced to ensure that each module has at least one valid input connection.

2) *CNN Module Parameter Encoding*: The 11th to 25th dimensions (15 dimensions in total) encode the parameters of the 3 independent CNN modules, with 5 consecutive integer dimensions assigned to each module:

$$\text{conv}_i = \{knum, ksize, kact, pt, ps\},$$

$$\begin{cases} knum = \text{Randint}(0, 7) \\ ksize = \text{Randint}(0, 3) \\ kact = \text{Randint}(0, 7) \\ pt = \text{Randint}(0, 2) \\ ps = \text{Randint}(0, 3) \end{cases} \quad (3)$$

The corresponding mapping rules are defined as follows:
 $knum$: Actual number of output channels = $16 \times (knum + 1)$
 $ksize$: Actual 1D convolutional kernel length = $2 \times ksize + 3$
 $kact$: Activation function mapping:

$$kact = \begin{cases} 0 : \text{Softplus} \\ 1 : \text{Softsign} \\ 2 : \text{ELU} \\ 3 : \text{Softmax} \\ 4 : \text{Sigmoid} \\ 5 : \text{Tanh} \\ 6 : \text{ReLU} \\ 7 : \text{Linear} \end{cases} \quad (4)$$

pt : Pooling type mapping:

$$pt = \begin{cases} 0 : \text{MaxPooling} \\ 1 : \text{AvgPooling} \\ 2 : \text{No pooling layer} \end{cases} \quad (5)$$

ps : If $pt \neq 2$, actual pooling kernel length = $2 \times ps + 3$.

3) *BiLSTM Module Parameter Encoding*: The 26th to 35th dimensions (10 dimensions in total) encode the parameters of the 2 independent BiLSTM modules, with 5 consecutive integer dimensions assigned to each module:

$$\text{bilstm}_j = \{hnum, lnum, afunc, ptype, drop\},$$

$$\begin{cases} hnum = \text{Randint}(0, 7) \\ lnum = \text{Randint}(0, 3) \\ afunc = \text{Randint}(0, 7) \\ ptype = \text{Randint}(0, 2) \\ drop = \text{Randint}(0, 3) \end{cases} \quad (6)$$

where $hnum$ corresponds to the actual hidden size per direction $16 \times (hnum + 1)$, $lnum$ corresponds to the actual number of stacked layers $lnum + 1$, $afunc$ follows the same activation mapping as CNN modules, $ptype$ follows the same pooling mapping as CNN modules, and $drop$ corresponds to the actual dropout probability $drop \times 0.1$.

4) *Training Hyperparameter Encoding*: The final 4 dimensions (36th to 39th) encode the key training hyperparameters:

$$\mathbf{x}_{\text{train}} = \{bs, opt, lr, reg\}, \begin{cases} bs = \text{Randint}(0, 3) \\ opt = \text{Randint}(0, 3) \\ lr \in [10^{-4}, 10^{-2}] \\ reg = \text{Randint}(0, 3) \end{cases} \quad (7)$$

where bs corresponds to the actual batch size $32 \times (bs + 1)$, and lr directly encodes the initial learning rate used in the training process. The optimizer mapping is:

$$opt = \begin{cases} 0 : \text{SGD} \\ 1 : \text{Adam} \\ 2 : \text{AdaDelta} \\ 3 : \text{RMSprop} \end{cases} \quad (8)$$

Algorithm 1 Overall Optimization Workflow of MoACB-WSF

```

1: Input: Population size  $N$ , maximum generation  $T$ ,
   crossover probability  $p_c$ , mutation probability  $p_m$ , training
   set  $\mathcal{D}_{train}$ , validation set  $\mathcal{D}_{val}$ 
2: Output: Final Pareto optimal set from external archive
    $A_E$ 
3:  $P_0 \leftarrow \text{InitializePopulation}(N)$ 
4:  $A_E \leftarrow \emptyset$ 
5: Evaluate bi-objective fitness  $(f_1, f_2)$  for all individuals in
    $P_0$ 
6: Update  $A_E$  with non-dominated solutions from  $P_0$ 
7:  $t \leftarrow 0$ 
8: while  $t < T$  do
9:   Perform fast non-dominated sorting and crowding distance
   calculation on  $P_t$ 
10:  Select parent individuals via binary tournament selection
11:   $Q_t \leftarrow \text{Crossover}(\text{parent population}, p_c)$ 
12:   $Q_t \leftarrow \text{Mutation}(Q_t, p_m)$ 
13:  Evaluate bi-objective fitness of all individuals in  $Q_t$ 
14:   $R_t \leftarrow P_t \cup Q_t$ 
15:  Perform fast non-dominated sorting on  $R_t$ 
16:  Update  $A_E$  with non-dominated solutions from  $R_t$ 
17:  Select top  $N$  individuals from  $R_t$  via elitist selection
   to form  $P_{t+1}$ 
18:   $t \leftarrow t + 1$ 
19: end while
20: Extract final Pareto optimal set from converged  $A_E$ 
21: return Final Pareto optimal set from  $A_E$ 

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The regularization mapping is defined as:

$$reg = \begin{cases} 0 : & \text{No regularization} \\ 1 : & \text{L1 regularization} \\ 2 : & \text{L2 regularization} \\ 3 : & \text{L1\&L2 regularization} \end{cases} \quad (9)$$

C. Customized NSGA-II Evolutionary Process

NSGA-II is adopted as the core multi-objective optimization framework, with customized evolutionary operators designed for the hybrid encoding structure of the CNN-BiLSTM model. The initial population is constructed via constrained random initialization, ensuring each individual satisfies both structural connectivity and parameter bound constraints, as detailed in Algorithm 2.

In each generation, fast non-dominated sorting partitions the population into hierarchical Pareto fronts based on dominance relationships. A solution x_1 dominates x_2 if and only if $f_k(x_1) \leq f_k(x_2)$ for all $k \in \{1, 2\}$ with at least one strict inequality, where f_1 denotes the validation RMSE and f_2 the total number of trainable parameters, both to be minimized.

To preserve population diversity and prevent premature convergence, the crowding distance for each individual is computed as:

$$d_i = \sum_{m=1}^2 \frac{f_m^{(i+1)} - f_m^{(i-1)}}{f_m^{\max} - f_m^{\min}}, \quad (10)$$

Algorithm 2 Population Initialisation

```

1: Input: The population size  $N$ 
2: Output: The initialised population  $P_0$ 
3:  $P_0 \leftarrow \emptyset$ 
4: for  $it \leftarrow 1$  to  $N$  do
5:    $I \leftarrow$  Initialise an empty array for individual encoding
6:   setting  $\leftarrow$  Randomly generate 4-integer array
7:    $I \leftarrow I \cup \text{setting}$ 
8:   topo  $\leftarrow$  Randomly generate 10-bit binary array
9:    $I \leftarrow I \cup \text{topo}$ 
10:  for  $j \leftarrow 1$  to 3 do
11:    cnn $_j \leftarrow$  Randomly generate 5-integer array
12:     $I \leftarrow I \cup \text{cnn}_j$ 
13:  end for
14:  for  $j \leftarrow 1$  to 2 do
15:    lstm $_j \leftarrow$  Randomly generate 5-integer array
16:     $I \leftarrow I \cup \text{lstm}_j$ 
17:  end for
18:   $P_0 \leftarrow P_0 \cup I$ 
19: end for
20: return  $P_0$ 

```

where boundary individuals are assigned infinite crowding distance to guarantee retention of elite solutions.

Parent selection is performed via binary tournament selection based on Pareto rank and crowding distance. A segment-preserving single-point crossover operator is designed to maintain the structural integrity of each encoding segment, with dedicated handling rules for the topology region. Fig. 4 depicts the detailed mechanism of the proposed crossover operation.

A variable-region mutation operator is further applied to enhance exploratory capability. In each mutation event, one of three encoding regions—topology, network module, or training hyperparameter—is randomly selected for regeneration, ensuring the mutated individual remains structurally valid. Fig. 5 depicts the detailed mechanism of the customized variable-region mutation.

An external archive A_E is maintained throughout evolution to store all non-dominated solutions. In each generation, the archive is updated by incorporating newly discovered non-dominated solutions and removing any dominated entries. When the archive size exceeds N , truncation based on crowding distance is applied to preserve solution diversity.

The next-generation population is determined via elitist environmental selection from the combined parent-offspring pool, prioritizing lower Pareto rank and, among equal-rank individuals, larger crowding distance. This mechanism ensures simultaneous convergence toward the true Pareto front and adequate coverage of the trade-off solution space.

D. Bi-Objective Fitness Evaluation

We formulate the automated design of hybrid CNN-BiLSTM models for wind speed forecasting as a bi-objective minimization problem:

$$\min_{\mathbf{x} \in \mathcal{F}} \mathbf{F}(\mathbf{x}) = [f_1(\mathbf{x}), f_2(\mathbf{x})]^\top \quad (11)$$

Segment-Preserving Single-Point Crossover

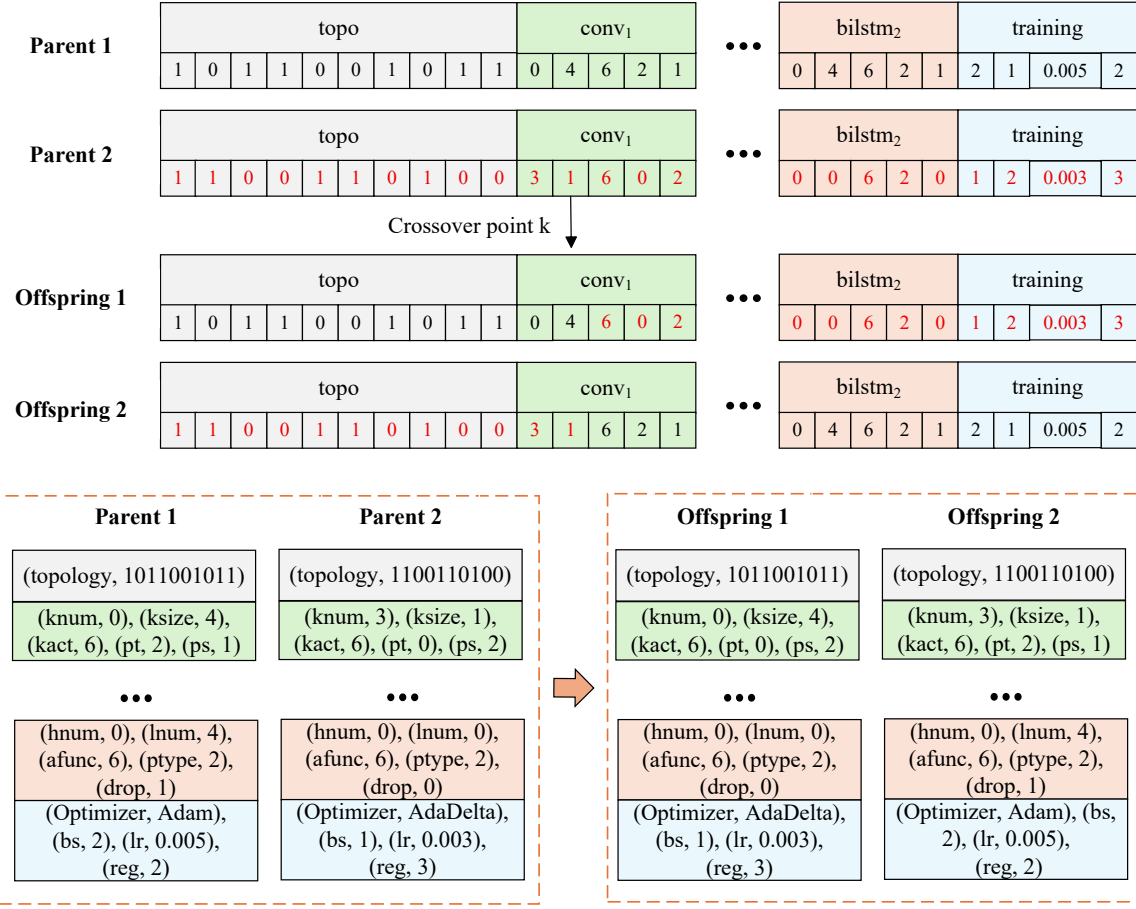


Fig. 4. Illustration of the segment-preserving single-point crossover operator.

Variable Region Mutation

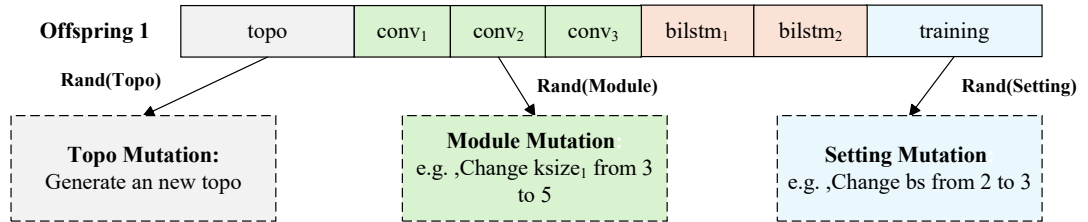


Fig. 5. Illustration of the customized variable-region mutation operator.

where the two objective functions are formally defined as follows:

$$f_1(\mathbf{x}) = \text{RMSE}(\mathbf{x}) = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i(\mathbf{x}))^2} \quad (12a)$$

$$f_2(\mathbf{x}) = \mathcal{C}(\mathbf{x}) = \sum_{i=1}^L \text{Param}_i(\mathbf{x}) \quad (12b)$$

Here, $f_1(\mathbf{x})$ denotes the root mean square error (RMSE) on the validation set, which quantifies the prediction accuracy of the model; $f_2(\mathbf{x})$ represents the total number of trainable parameters, which measures the model complexity; y_i and $\hat{y}_i(\mathbf{x})$ are the actual and predicted wind speed values, respectively; N is the number of validation samples; and $\text{Param}_i(\mathbf{x})$ is the number of trainable parameters in the i -th module of the model.

Algorithm 3 Segment-Preserving Single-Point Crossover Operation

```

1: Input: Parent population  $P$ , crossover probability  $p_c$ , population size  $N$ ,
2:   total encoding length  $L$ , topology segment length  $L_t$ 
3: Output: Offspring population  $Q$ 
4:  $Q \leftarrow \emptyset$ 
5: while  $|Q| < N$  do
6:   Randomly select two distinct individuals  $par_1, par_2$  from  $P$ 
7:    $r \sim \mathcal{U}(0, 1)$ 
8:   if  $r < p_c$  then
9:     Randomly select crossover point  $k \in \{0, 1, \dots, L-1\}$ 
10:    if  $k < L_t$  then
11:       $child_1 \leftarrow par_1$ 
12:       $child_2 \leftarrow par_2$ 
13:      Replace topology of  $child_1$  with a valid  $L_t$ -bit binary topology
14:      Replace topology of  $child_2$  with a new valid  $L_t$ -bit binary topology
15:    else Crossover in module/training segments
16:      Split  $par_1$  into  $p_1$  and  $p_2$ 
17:      Split  $par_2$  into  $p_3$  and  $p_4$ 
18:       $child_1 \leftarrow p_1 \cup p_4$ 
19:       $child_2 \leftarrow p_3 \cup p_2$ 
20:    end if
21:  else
22:     $child_1 \leftarrow par_1, child_2 \leftarrow par_2$ 
23:  end if
24:   $Q \leftarrow Q \cup \{child_1, child_2\}$ 
25: end while
26: return  $Q$ 

```

Algorithm 4 Variable-Region Mutation Operation**Require:** Offspring population Q , mutation probability p_m **Ensure:** Mutated population Q_M

```

1:  $Q_M \leftarrow \emptyset$ 
2: for each individual  $x \in Q$  do
3:    $x_{tmp} \leftarrow x$ 
4:    $r \leftarrow \text{Random}(0, 1)$ 
5:   if  $r < p_m$  then
6:      $R \leftarrow \text{RandChoice}(\{\text{Topo}, \text{Module}, \text{Setting}\})$ 
7:     if  $R = \text{Topo}$  then
8:       new_topo  $\leftarrow$  random 10-bit binary vector
9:       Replace topo segment in  $x_{tmp}$  with new_topo
10:    else
11:      Update  $x_{tmp}$  using a valid parameter in region  $R$ 
12:    end if
13:  end if
14:   $Q_M \leftarrow Q_M \cup \{x_{tmp}\}$ 
15: end for
16: return  $Q_M$ 

```

Algorithm 5 Bi-Objective Fitness Evaluation

```

1: Input: Encoded individual  $x$ , training set  $\mathcal{D}_{train}$ , validation set  $\mathcal{D}_{val}$ 
2: Output: Bi-objective fitness values  $(f_1, f_2)$ 
3: Decode the topology, module parameters, and training hyperparameters from  $x$ 
4: Dynamically construct the hybrid CNN-BiLSTM model  $M$  with the decoded structure
5: Compute the total number of trainable parameters as  $f_2 = \mathcal{C}(x)$ 
6: Train model  $M$  on  $\mathcal{D}_{train}$  using the decoded hyperparameters, with early stopping to prevent overfitting
7: Generate wind speed predictions on the validation set  $\mathcal{D}_{val}$ 
8: Calculate the validation RMSE as  $f_1 = \text{RMSE}(x)$ 
9: return  $(f_1, f_2)$ 

```

The detailed procedure of bi-objective fitness evaluation is outlined in Algorithm 5.

E. External Archive Update and Best-Compromise Solution Selection

The external archive A_E is designed to permanently preserve all non-dominated solutions generated during the entire evolutionary process, which represent the elite hybrid CNN-BiLSTM models balancing prediction accuracy and model complexity. The archive A_E is updated iteratively following three core rules: 1. First, all non-dominated solutions identified from the combined parent-offspring population R_t are added to A_E ; 2. Subsequently, any existing solutions in A_E that are dominated by the newly added solutions are removed to strictly maintain the non-dominated property of the archive; 3. Finally, if the size of the archive exceeds the predefined population size N (i.e., $|A_E| > N$), A_E is truncated by eliminating individuals with the smallest crowding distances, thus preserving the diversity of the archived solutions in the objective space.

Upon the termination of the evolutionary process, the best-compromise solution is selected from the final converged archive A_E based on the normalized Euclidean distance to the ideal point in the bi-objective space. The ideal point is defined as the hypothetical solution with the minimum values of both objectives. Formally, the best-compromise solution x^* is determined by:

$$x^* = \arg \min_{x \in A_E} \sqrt{\left(\frac{f_1(x) - f_1^{\min}}{f_1^{\max} - f_1^{\min}}\right)^2 + \left(\frac{f_2(x) - f_2^{\min}}{f_2^{\max} - f_2^{\min}}\right)^2} \quad (13)$$

where $f_1^{\min}$, $f_1^{\max}$, $f_2^{\min}$, and $f_2^{\max}$ denote the minimum and maximum values of the two objectives within the final archive A_E , respectively. The normalization of objectives ensures that both prediction accuracy and model complexity contribute equally to the distance metric. The best-compromise hybrid CNN-BiLSTM model corresponding to x^* is then fully trained on the entire dataset for final deployment.

IV. EXPERIMENTAL RESULTS

This section presents a systematic experimental evaluation of the proposed MoACB-WSF model using three real-world wind farm datasets. First, we introduce the datasets used and the corresponding data preprocessing pipeline. Then, we define the evaluation metrics and detail the experimental setup. Subsequently, we conduct comprehensive comparative analysis between the proposed model and representative state-of-the-art manually designed models. Finally, we present an in-depth analysis of the multi-objective optimization mechanism, generalization robustness, and computational overhead of the proposed automated modeling framework.

A. Datasets

To validate the prediction accuracy, generalization capability, and adaptability to different temporal resolutions of the proposed model, three real-world benchmark datasets from operational wind farms are employed. The key characteristics of the datasets are summarized in Table I, and detailed descriptions are as follows:

1 Sotavento Wind Farm 10-minute Dataset

This dataset was acquired from the SCADA system of the Sotavento Wind Farm in Galicia, Spain, with a 10-minute sampling interval [33], [34]. The input features include wind speed, wind direction, theoretical power curve, and low-voltage active power, with wind speed as the prediction target. The dataset is partitioned into training (70%), validation (15%), and test (15%) sets via time-series stratified partitioning to avoid data leakage.

2 Turkish Wind Farm 10-minute Dataset

This dataset is from an onshore wind farm in western Turkey, with a 10-minute sampling interval and the same four input features as the Sotavento dataset. It contains 1132 valid samples covering one week, capturing the diurnal variability of wind speed in this region. The same 70%/15%/15% partitioning strategy is used for consistency.

3 Turkish Wind Farm 1-hour Dataset

Derived from the same Turkish wind farm with a 1-hour sampling interval, this dataset contains 1000 valid samples over six weeks. The coarser temporal resolution increases the non-stationarity of the wind speed series, making prediction more challenging. The same partitioning strategy is maintained for cross-resolution comparison.

The key statistical characteristics (max, median, min, mean, standard deviation) of wind speed in each dataset are summarized in Tables II, III, and IV, and the wind speed time series are visualized in Figs. 6, 7, and 8.

TABLE II
STATISTICAL CHARACTERISTICS OF THE SOTAVENTO 10-MINUTE DATASET (UNITS: M/S)

Dataset	Max	Median	Min	Mean	St.d
Entire dataset	16.29	6.84	0.29	7.28	3.87
Training dataset	16.29	6.41	0.29	6.63	3.77
Validation dataset	16.05	11.59	2.95	10.65	3.11
Test dataset	15.36	6.62	1.67	6.93	3.26

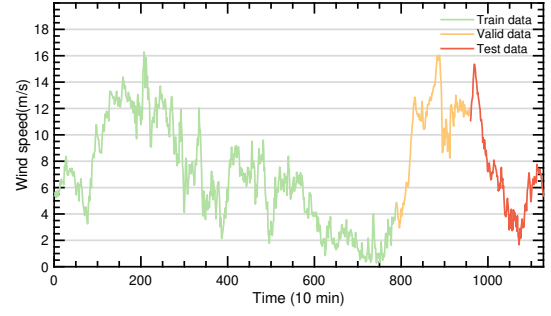


Fig. 6. Actual wind speed data of the Sotavento 10-minute dataset (Train/Val/Test split).

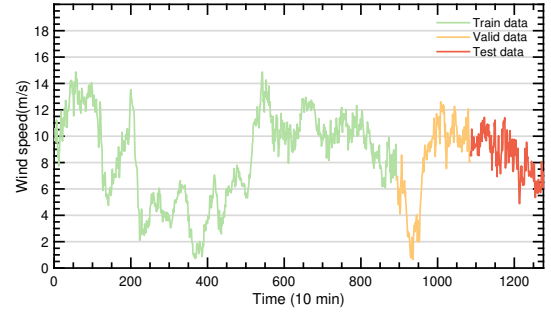


Fig. 7. Actual wind speed data of the Turkish 10-minute dataset (Train/Val/Test split).

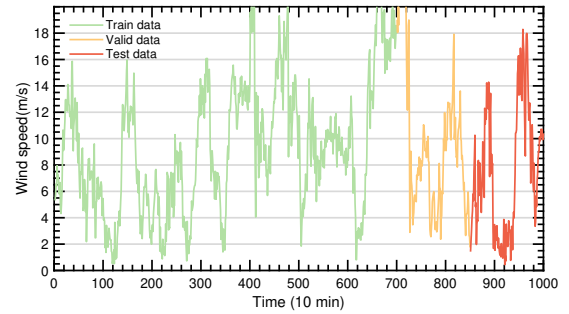


Fig. 8. Actual wind speed data of the Turkish 1-hour dataset (Train/Val/Test split).

B. Data Preprocessing

A unified and rigorous data preprocessing pipeline is implemented for all datasets to ensure data quality and experimental fairness:

1. **Quality Control:** Invalid samples (e.g., negative wind speed) are removed, missing values are supplemented via linear interpolation, and outliers are eliminated using the 3σ criterion.
2. **Normalization:** All input features are normalized to the range $[0,1]$ using Min-Max scaling to avoid feature scaling bias:

$$x_{\text{norm}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \quad (14)$$

TABLE I
STATISTICAL CHARACTERISTICS OF THE ADOPTED DATASETS

Dataset Name	Sampling Interval	Total Samples	Number of Features	Time Span
Sotavento Wind Farm	10 minutes	1128	4	1 week
Turkish Wind Farm (10-min)	10 minutes	1132	4	1 week
Turkish Wind Farm (1-hour)	1 hour	1000	4	6 weeks

TABLE III
STATISTICAL CHARACTERISTICS OF THE TURKISH 10-MINUTE DATASET
(UNITS: M/S)

Dataset	Max	Median	Min	Mean	St.d
Entire dataset	14.87	9.04	0.65	8.36	3.14
Training dataset	14.87	9.13	0.73	8.43	3.37
Validation dataset	12.61	9.18	0.65	7.94	3.22
Test dataset	11.43	8.69	4.88	8.49	1.57

TABLE IV
STATISTICAL CHARACTERISTICS OF THE TURKISH 1-HOUR DATASET
(UNITS: M/S)

Dataset	Max	Median	Min	Mean	St.d
Entire dataset	24.09	8.62	0.44	9.09	5.04
Training dataset	24.09	8.83	0.52	9.30	4.91
Validation dataset	24.04	8.58	2.37	9.70	5.55
Test dataset	18.29	6.79	0.44	7.47	4.84

where $x_{\min}$ and $x_{\max}$ are the minimum and maximum values of the feature, respectively.

3. Time-Series Partitioning: The dataset is partitioned into training (70%), validation (15%), and test (15%) sets in chronological order to preserve the temporal distribution characteristics of wind speed data and avoid data leakage.

C. Metrics

To quantitatively evaluate the performance of wind speed prediction models from both accuracy and model complexity perspectives, five metrics are adopted in this work. The first four metrics, i.e., Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Mean Absolute Percentage Error (MAPE), and Pearson Correlation Coefficient (R), are specifically used to assess the wind speed prediction accuracy, while the last metric, Number of Trainable Parameters (Params), is employed to measure the model's computational complexity and lightweightness. For MAE, RMSE, and MAPE, lower values indicate better prediction accuracy. A value of R close to 1 signifies a stronger linear correlation between the predicted and actual wind speed sequences, reflecting the model's ability to capture the temporal variation trend of wind speed. The mathematical definitions of these four prediction accuracy metrics are given as follows:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (15)$$

$$\text{MAPE} = \frac{100}{N} \sum_{i=1}^N \frac{|y_i - \hat{y}_i|}{y_i}$$

$$R = \frac{\sum_{i=1}^N (y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_{i=1}^N (y_i - \bar{y})^2} \sqrt{\sum_{i=1}^N (\hat{y}_i - \bar{\hat{y}})^2}}$$

where N denotes the total number of wind speed samples, y_i and $\hat{y}_i$ represent the i -th actual and predicted wind speed values respectively, and $\bar{y}$ and $\bar{\hat{y}}$ are the mean values of the actual and predicted wind speed sequences.

The number of trainable parameters quantifies the model's computational complexity and memory footprint, which is a fundamental concern in deep learning [35]. Fewer parameters correspond to a more lightweight model with lower memory consumption and faster inference speed, which is critical for real-time wind speed prediction deployed on edge devices and wind farm SCADA systems. Its mathematical expression is:

$$\text{Params} = \sum_{i=1}^L \text{Param}_i \quad (16)$$

where L denotes the total number of layers in the model, and Param_i represents the number of trainable parameters in the i -th layer.

D. Compared With Other Wind Speed Prediction Methods

This section presents comprehensive experimental results and comparisons with state-of-the-art wind speed prediction models to validate the effectiveness, efficiency, and generalization capability of the proposed MoACB-WSF framework. All experiments are implemented on three real-world wind speed datasets, namely the Sotavento 10-min dataset, the Turkish 10-min SCADA dataset, and the Turkish 1-h SCADA dataset. Six representative manually designed models with carefully hand-tuned hyperparameters from their original publications are selected as baseline methods, including CNN [36], WP-Former [4], EEMD-BiLSTM [2], LEMQ-MSDCN [16], BiLSTM [37], and LSTM-Transformer [38]; additionally, two leading neuroevolution-based approaches, i.e., GA-GLSTM

[39] and Mr-Net [24], are also adopted for comprehensive comparison. The prediction performance is evaluated by five widely accepted metrics: mean absolute error (MAE), root mean square error (RMSE), mean absolute percentage error (MAPE), correlation coefficient (R), and the number of trainable parameters (Params) for quantifying model complexity.

For the multi-objective architecture optimization, the NSGA-II algorithm is configured with a population size of 60 and a maximum of 30 generations, using crossover and mutation probabilities of 0.8 and 0.5, respectively. All optimizations and validations are performed for one-step-ahead wind speed time series prediction with a prediction step size of 1. During the search stage, each candidate architecture is trained for 20 epochs with an early stopping patience of 5 epochs, while the final optimized model undergoes comprehensive retraining for 200 epochs with an extended patience of 20 epochs. The best trade-off hybrid encoding vectors of MoACB-WSF for different datasets are presented in Table V.

All experiments are conducted on a personal laptop equipped with an Intel Core i7-13700H CPU, an NVIDIA GeForce RTX 4070 Laptop GPU (8 GB dedicated graphics memory), and 16 GB RAM. All models are implemented using Python 3.9.25 and PyTorch 2.5.1. To eliminate the impact of random parameter initialization, all models are independently trained and evaluated five times under identical experimental settings. The optimal result from these five independent runs is selected as the final reported performance for each model.

Table VI presents the quantitative comparison results on the Sotavento 10-min dataset, with optimal metric values highlighted in bold. The best trade-off solution of the proposed MoACB-WSF achieves an MAE of 0.4215 m/s, an RMSE of 0.5052 m/s, and a correlation coefficient R of 0.9871 using only 16.9×10^3 trainable parameters. The best performance solution attains the optimal prediction metrics, with an MAE of 0.3924 m/s, an RMSE of 0.5035 m/s, and an R of 0.9875. Compared with the manually designed EEMD-BiLSTM baseline, MoACB-WSF reduces model complexity by 91.3% while maintaining comparable prediction accuracy, demonstrating the effectiveness of joint neural architecture and hyperparameter optimization for lightweight model design.

Table VII reports the experimental results on the Turkish 10-min SCADA dataset. Consistent with the Sotavento results, the proposed MoACB-WSF outperforms all hand-crafted and neuroevolution-based baselines. Specifically, the best trade-off solution achieves an MAE of 0.6605 m/s, an RMSE of 0.8428 m/s, a MAPE of 8.05%, and an R of 0.8554 with only 14.5×10^3 parameters. The best performance solution further improves prediction accuracy, yielding the lowest MAE of 0.6598 m/s and RMSE of 0.8243 m/s. These results validate the strong generalization capability of the proposed framework across diverse wind farm conditions.

Table VIII illustrates the performance on the Turkish 1-hour dataset, which exhibits higher non-stationarity and prediction difficulty due to the longer sampling interval. Even in this challenging scenario, MoACB-WSF achieves state-of-the-art performance. The best trade-off solution obtains the lowest MAE of 1.4470 m/s with only 13.6×10^3 parameters, while the best performance solution achieves the optimal RMSE of

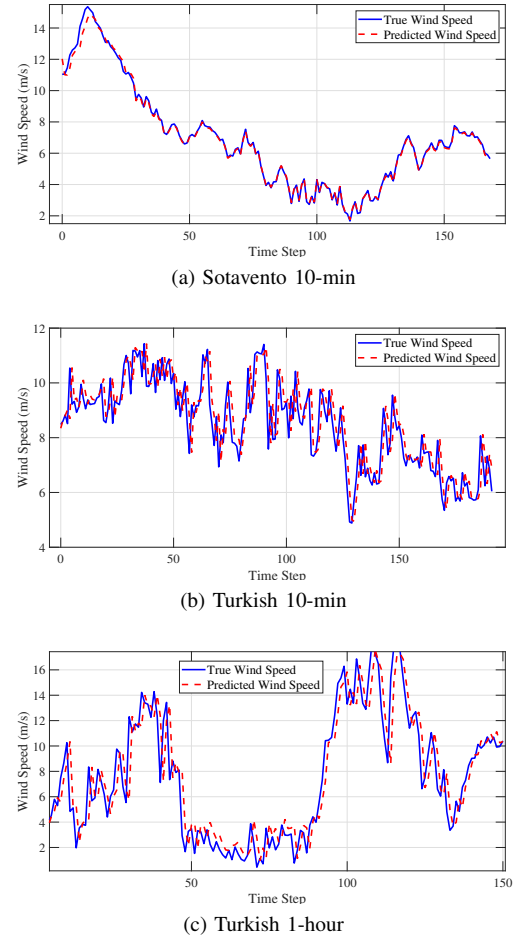


Fig. 9. Predicted and measured wind speed profiles of the best trade-off solution on the three test datasets: (a) Sotavento 10-min, (b) Turkish 10-min, and (c) Turkish 1-hour.

1.9322 m/s and the highest R of 0.9163. These results further verify the superiority of MoACB-WSF for long-term wind speed forecasting.

Fig. 9 compares the predicted and measured wind speed profiles of the best trade-off solution across all datasets. The predicted profiles closely match the measured values under both short-term and long-term forecasting settings, with no noticeable systematic bias. These qualitative results further confirm that MoACB-WSF effectively models the nonlinear dynamics of wind speed while maintaining an ultra-compact structure, which is advantageous for real-time edge deployment in wind energy systems.

Fig. 10 illustrates the evolutionary trajectories of the Pareto-optimal fronts (POFs) at generations $t = 1$, $t = 5$, $t = 10$, $t = 20$, and $t = 30$ on the Sotavento 10-min and Turkish 1-hour datasets, which visualizes the multi-objective search process toward optimal configurations of the hybrid CNN-BiLSTM architecture, module parameters, and training hyperparameters. As can be observed, the POFs of both datasets gradually converge toward the lower-left ideal region through the iterative evolution of the customized NSGA-II. Specifically, the initial POF at $t = 1$ exhibits relatively large prediction error and model complexity. As evolution proceeds,

TABLE V
BEST TRADE-OFF HYBRID ENCODING VECTORS OBTAINED BY THE PROPOSED MoACB-WSF

Dataset	<i>Topo</i>	CNN	BiLSTM	Setting
Sotavento 10-min Dataset	[1,1,1,1,0,0,0,0,0]	[0,0,1,0,1] [1,3,4,1,1] [3,0,3,1,1]	[0,0,7,1,0] [0,0,0,1,2]	[0, 0, 0.0072, 3]
Turkish 10-min Dataset	[1,0,0,0,1,1,0,0,0,1]	[0,1,7,1,3] [0,0,6,0,1] [2,0,6,1,3]	[0,0,2,2,2] [0,0,7,1,2]	[0, 3, 0.0002, 3]
Turkish 1-hour Dataset	[1,1,0,1,0,1,0,0,0,0]	[0,2,2,0,3] [0,2,7,0,1] [2,0,5,0,3]	[0,0,6,0,2] [0,0,2,0,1]	[3, 0, 0.0053, 1]

TABLE VI
PERFORMANCE COMPARISON ON THE SOTAVENTO 10-MIN DATASET

	Model	MAE	RMSE	MAPE (%)	<i>R</i>	Params ($\times 10^3$)
Manually designed models	CNN [36]	0.5169 (8)	0.6722 (8)	7.25 (1)	0.9835 (6)	273.0 (10)
	WPFormer [4]	0.4128 (3)	0.5245 (5)	8.27 (4)	0.9866 (4)	28.9 (4)
	EEMD-BiLSTM [2]	0.4079 (2)	0.5089 (3)	8.33 (6)	0.9704 (10)	194.3 (8)
	LEMQ-MSDCN [16]	0.5834 (10)	0.7315 (9)	10.89 (10)	0.9737 (9)	43.3 (5)
	BiLSTM [37]	0.4262 (6)	0.5329 (6)	8.36 (7)	0.9867 (3)	135.2 (7)
	LSTM-Transformer [38]	0.5461 (9)	0.7474 (10)	9.88 (9)	0.9743 (8)	95.5 (6)
Neuroevolution-based models	GA-GLSTM [39]	0.4217 (5)	0.5222 (4)	8.31 (5)	0.9865 (5)	210.2 (9)
	Mr-Net [24]	0.4675 (7)	0.5840 (7)	9.29 (8)	0.9833 (7)	24.9 (3)
Ours	MoACB-WSF (Best Trade-off)	0.4215 (4)	0.5052 (2)	8.12 (3)	0.9871 (2)	16.9 (1)
	MoACB-WSF (Best Performance)	0.3924 (1)	0.5035 (1)	7.99 (2)	0.9875 (1)	24.4 (2)

TABLE VII
PERFORMANCE COMPARISON ON THE TURKISH 10-MIN DATASET

	Model	MAE	RMSE	MAPE (%)	<i>R</i>	Params ($\times 10^3$)
Manually designed models	CNN [36]	0.6851 (8)	0.8649 (7)	10.92 (10)	0.9402 (1)	273.0 (9)
	WPFormer [4]	0.6688 (4)	0.8472 (4)	8.08 (3)	0.8432 (6)	28.9 (2)
	EEMD-BiLSTM [2]	0.6673 (3)	0.8469 (3)	8.07 (2)	0.8924 (2)	194.3 (8)
	LEMQ-MSDCN [16]	0.7096 (9)	0.8778 (8)	8.66 (8)	0.8304 (8)	43.3 (5)
	BiLSTM [37]	0.6726 (5)	0.8559 (5)	8.18 (7)	0.8453 (5)	135.2 (7)
	LSTM-Transformer [38]	0.7679 (10)	0.9925 (10)	9.57 (9)	0.8016 (10)	95.5 (6)
Neuroevolution-based models	GA-GLSTM [39]	0.6754 (6)	0.8587 (6)	8.16 (6)	0.8394 (7)	955.4 (10)
	Mr-Net [24]	0.6811 (7)	0.8873 (9)	8.12 (4)	0.8263 (9)	32.9 (3)
Ours	MoACB-WSF (Best Trade-off)	0.6605 (2)	0.8428 (2)	8.05 (1)	0.8554 (4)	14.5 (1)
	MoACB-WSF (Best Performance)	0.6598 (1)	0.8243 (1)	8.15 (5)	0.8616 (3)	34.6 (4)

TABLE VIII
PERFORMANCE COMPARISON ON THE TURKISH 1-HOUR DATASET

	Model	MAE	RMSE	MAPE (%)	R	Params ($\times 10^3$)
Manually designed models	CNN [36]	1.5851 (8)	2.0925 (8)	34.33 (3)	0.9068 (4)	273.0 (10)
	WPFormer [4]	1.5132 (6)	2.0424 (5)	37.9484 (7)	0.9050 (6)	28.9 (2)
	EEMD-BiLSTM [2]	1.4879 (4)	1.9762 (3)	51.55 (10)	0.9156 (2)	194.3 (8)
	LEMQ-MSDCN [16]	1.5066 (5)	2.0532 (7)	34.18 (2)	0.9020 (7)	43.3 (5)
	BiLSTM [37]	1.5163 (7)	2.0512 (6)	36.51 (6)	0.9051 (5)	135.2 (7)
	LSTM-Transformer [38]	1.6909 (9)	2.2280 (10)	39.79 (8)	0.8870 (9)	95.5 (6)
Neuroevolution-based models	GA-GLSTM [39]	1.4865 (3)	2.0033 (4)	30.04 (1)	0.8703 (10)	243.7 (9)
	Mr-Net [24]	1.7182 (10)	2.1951 (9)	48.67 (9)	0.8894 (8)	42.1 (4)
Ours	MoACB-WSF (Best Trade-off)	1.4470 (1)	1.9660 (2)	35.09 (5)	0.9137 (3)	13.6 (1)
	MoACB-WSF (Best Performance)	1.4492 (2)	1.9322 (1)	35.04 (4)	0.9163 (1)	31.1 (3)

the non-dominated solutions obtained at $t = 30$ exhibit clear dominance over those from earlier generations, indicating the strong global exploration and local exploitation capabilities of the proposed framework. The consistent evolutionary tendencies verify that the hybrid encoding strategy and tailored evolutionary operators enable the customized NSGA-II to generate well-distributed and well-converged POFs. Accordingly, MoACB-WSF can automatically identify lightweight yet high-precision wind speed prediction models while achieving a favorable balance between forecasting accuracy and model complexity, which effectively reduces the heavy dependence on manual expert experience and empirical trial-and-error in practical wind farm applications.

Tables IX and X summarize the experimental results of the best-performance and best trade-off solutions on the Sotavento 10-min dataset across the five independent runs, respectively. The corresponding results on the Turkish 1-hour dataset are presented in Tables XI and XII. The distribution of the final Pareto-optimal external archive obtained from the five independent runs is visualized in Fig. 11.

The quantitative results in the above tables demonstrate that MoACB-WSF exhibits strong stability, with negligible fluctuations in all evaluation metrics across the repeated experiments. Compared with the best-performance solution, the best trade-off solution achieves a substantial reduction in model parameters with only a minor degradation in prediction accuracy, which validates the effectiveness of the multi-objective optimization mechanism. As shown in Fig. 11, the Pareto fronts obtained from the five independent runs are highly consistent and well-converged. These observations confirm that the proposed framework is robust to random initialization and reliably produces high-quality lightweight models, thereby ensuring its practical applicability and reproducibility in real-world wind farm engineering scenarios.

In summary, extensive experimental results demonstrate that the proposed MoACB-WSF outperforms state-of-the-art hand-crafted models and neuroevolution-based approaches in both prediction accuracy and model lightweight. By jointly optimizing network topology, module parameters and training hyperparameters via multi-objective evolutionary algorithm,

TABLE IX
RESULTS OF MoACB-WSF (BEST PERFORMANCE) ON THE SOTAVENTO 10-MIN DATASET ACROSS FIVE INDEPENDENT RUNS

	MAE	RMSE	MAPE	R	Params ($\times 10^3$)
Run 1	0.4215	0.5432	8.42	0.9861	93.5
Run 2	0.3924	0.5035	7.99	0.9875	24.4
Run 3	0.4192	0.5405	8.37	0.9864	17.1
Run 4	0.4263	0.5298	8.58	0.9855	11.1
Run 5	0.4229	0.5651	8.46	0.9860	42.5

TABLE X
RESULTS OF MoACB-WSF (BEST TRADE-OFF) ON THE SOTAVENTO 10-MIN DATASET ACROSS FIVE INDEPENDENT RUNS

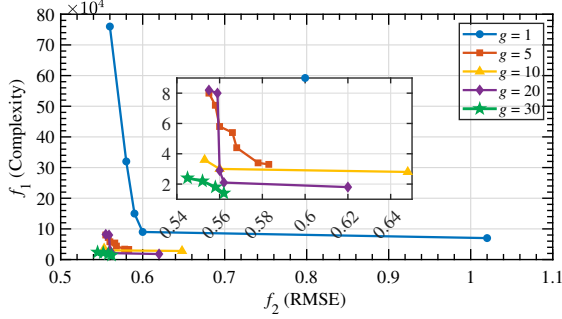
	MAE	RMSE	MAPE	R	Params ($\times 10^3$)
Run 1	0.4287	0.5523	8.60	0.9816	35.8
Run 2	0.4215	0.5052	8.12	0.9871	16.9
Run 3	0.4675	0.5501	8.55	0.9859	14.1
Run 4	0.4413	0.5594	8.78	0.9845	10.8
Run 5	0.4498	0.5732	8.98	0.9849	38.1

TABLE XI
RESULTS OF MoACB-WSF (BEST PERFORMANCE) ON THE TURKISH 1-HOUR DATASET ACROSS FIVE INDEPENDENT RUNS

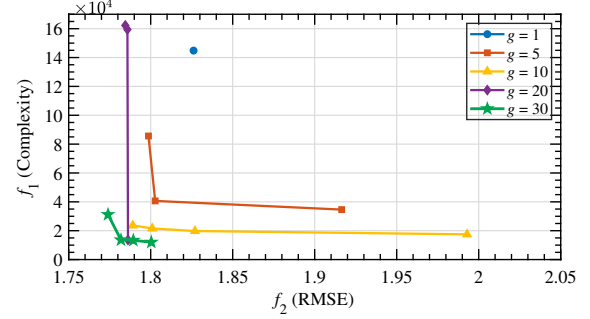
	MAE	RMSE	MAPE	R	Params ($\times 10^3$)
Run 1	1.4417	1.9312	35.88	0.9132	49.7
Run 2	1.4231	1.9291	35.33	0.9118	15.8
Run 3	1.4492	1.9322	36.04	0.9163	31.2
Run 4	1.4905	1.9403	37.26	0.9167	14.8
Run 5	1.4594	1.9654	36.72	0.9126	20.8

TABLE XII
RESULTS OF MoACB-WSF (BEST TRADE-OFF) ON THE TURKISH 1-HOUR DATASET ACROSS FIVE INDEPENDENT RUNS

	MAE	RMSE	MAPE	R	Params ($\times 10^3$)
Run 1	1.4250	1.9545	35.34	0.9098	16.5
Run 2	1.4410	1.9682	36.69	0.9122	11.4
Run 3	1.4470	1.9660	35.09	0.9137	13.6
Run 4	1.4895	1.9687	38.67	0.9131	12.3
Run 5	1.5012	1.9854	37.02	0.9091	12.5

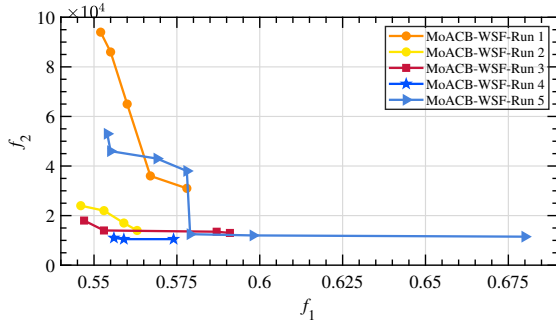


(a) Sotavento 10-min dataset

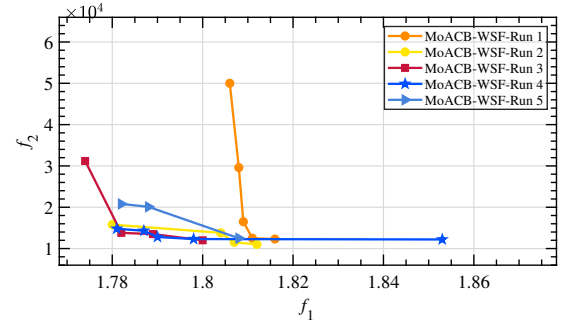


(b) Turkish 1-hour dataset

Fig. 10. Evolutionary trajectories of Pareto-optimal fronts across representative generations on two wind speed datasets: (a) Sotavento 10-min dataset, (b) Turkish 1-hour dataset, where $t = i$ denotes the i -th generation.



(a) Sotavento 10-min dataset



(b) Turkish 1-hour dataset

Fig. 11. Distribution of the Pareto-optimal external archive obtained from five independent runs on two wind speed datasets: (a) Sotavento 10-min dataset; (b) Turkish 1-hour dataset.

MoACB-WSF automatically constructs high-performance prediction models without manual expert intervention. Such advantages make the proposed framework a promising solution for reliable and real-time wind speed prediction in wind farms.

E. Discussion

From Tables VI, VII, and VIII, the proposed automated deep learning-driven wind speed forecasting framework, MoACB-WSF, demonstrates superior overall performance compared with state-of-the-art competitors in terms of prediction error metrics (MAE, RMSE, and MAPE), Pearson correlation coefficient R , and model lightweightness. This advantage stems from the NSGA-II-based multi-objective optimization mechanism, which autonomously explores high-performance hybrid CNN-BiLSTM architectures and effectively mitigates the inherent limitations of manually designed models, such as over-parameterization and heavy reliance on expert knowledge. By leveraging the powerful global search capability of the evolutionary algorithm, together with the proposed hybrid encoding strategy and customized evolutionary operators, MoACB-WSF achieves more competitive performance than conventional neuroevolution-based forecasting methods. Furthermore, the framework exhibits strong generalization across diverse wind farm datasets with varying sampling intervals and autonomously identifies models that optimally

balance accuracy and complexity, thereby facilitating practical deployment in real-world wind speed prediction applications.

The proposed MoACB-WSF has the following limitations. First, it searches a large hybrid space that encompasses both neural architectures and hyperparameters, rendering the optimization process computationally intensive and time-consuming. Second, as a purely data-driven deep learning method, MoACB-WSF offers limited interpretability and transparency, which complicates detailed analysis of its decision-making mechanisms for wind speed prediction. Third, the neural architecture search is performed entirely offline, restricting the model's adaptability to time-varying wind speed data that may exhibit distribution shifts.

To address these limitations, several promising directions will be pursued in future research. First, advanced multi-objective optimization techniques—such as multi-fidelity optimization, surrogate-assisted search, and state-of-the-art evolutionary algorithms—will be incorporated to reduce computational overhead and accelerate architecture search within large-scale hybrid spaces. Second, explainable artificial intelligence (XAI) methods, including gradient-based attribution analysis and attention visualization, will be integrated to enhance model interpretability and transparency, thereby revealing the intrinsic reasoning mechanisms underlying wind speed prediction. Third, an online adaptive neural architecture search framework will be developed to enable dynamic model updating, allowing

effective adaptation to evolving wind speed characteristics and potential distribution drifts in real-world scenarios. In addition, the model will be extended to multi-step-ahead forecasting via sequence-to-sequence architectures, and multi-source data fusion will be adopted to further improve robustness, ultimately promoting the practical implementation of the proposed method in industrial wind farm systems.

V. CONCLUSION

In this paper, we propose MoACB-WSF, a novel multi-objective automated hybrid deep learning framework tailored for wind speed forecasting. By integrating a CNN-BiLSTM backbone with an efficient fixed-length hybrid encoding mechanism, the proposed framework effectively captures the complex temporal dependencies inherent in wind speed series. We leverage the NSGA-II algorithm to jointly optimize both the neural network architecture and the training hyperparameters within a single unified search space, successfully overcoming the traditional reliance on labor-intensive, expert-driven manual model design. To validate the proposed method, extensive experiments were conducted across three real-world wind farm datasets, comparing MoACB-WSF against six representative manually designed models and two state-of-the-art neuroevolution baseline. The experimental results demonstrate the superior comprehensive performance of MoACB-WSF, confirming its ability to significantly enhance prediction precision while simultaneously maintaining exceptional model lightweightness. Ultimately, the proposed framework achieves an optimal accuracy-complexity trade-off, providing a highly practical, automated, and robust solution for real-time wind speed forecasting in modern wind farm systems.

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